

DBA4811 Individual Assignment 2

1. Problem Statement

The objective of this engagement is to build a predictive model that identifies customers who are likely to default on their next credit card payment. We compare four logistic-regression models, including baseline, stepwise selection, and regularized variants using the Area-Under-the-ROC-Curve (AUC) as the key evaluation metric, enabling the bank to proactively manage credit risk and optimise intervention strategies.

2. Feature Engineering

The following engineered features capture recent repayment behaviour across different time horizons (1 month and 3 months), customer characteristics and credit exposure

a) worsening_delay	- Shows increases in delays over the last three payments using columns <code>payment_status_1</code>, <code>payment_status_2</code>, <code>payment_status_3</code> - Returns 1 if payment status shows deterioration across any 2 chronological months, and 0 otherwise.
b) is_delay	- Captures any delays in recent repayment using columns <code>payment_status_1</code>, <code>payment_status_2</code>, <code>payment_status_3</code> - Returns 1 if there were delays in the past 3 months, and 0 otherwise.
c) age x credit_limit	- Interaction between <code>age</code> and <code>credit_limit</code> columns - Captures domain-specific consumer characteristics that are known to affect credit risk.
d) latest_debt_to_credit_ratio	- Measures client's debt exposure with $\frac{\text{most recent due amount}}{\text{credit limit}}$ using columns <code>payment_amt1</code> and <code>credit_limit</code> - The greater the proportion of debt amount to credit limit, the closer the value is to 1

3. Model Selection

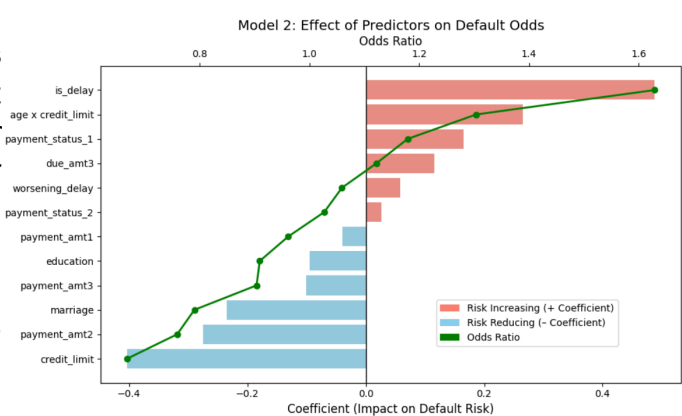
A total of six models were evaluated (see notebook), but only four key models are shown below for brevity.

	Model	Confusion Matrix	AUC Scores												
1	Baseline Logistic Regression	<table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td></tr><tr><td>0</td><td>0.80</td><td>0.86</td><td>0.83</td></tr><tr><td>1</td><td>0.54</td><td>0.42</td><td>0.47</td></tr></table>		precision	recall	f1-score	0	0.80	0.86	0.83	1	0.54	0.42	0.47	Train AUC: 0.74 Test AUC: 0.79 (0.785)
	precision	recall	f1-score												
0	0.80	0.86	0.83												
1	0.54	0.42	0.47												
2	Forward Stepwise Feature Selection with Balanced Class Weights (12 features)	<table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td></tr><tr><td>0</td><td>0.86</td><td>0.77</td><td>0.81</td></tr><tr><td>1</td><td>0.52</td><td>0.67</td><td>0.59</td></tr></table>		precision	recall	f1-score	0	0.86	0.77	0.81	1	0.52	0.67	0.59	Train AUC: 0.73 Test AUC: 0.79 (0.788)
	precision	recall	f1-score												
0	0.86	0.77	0.81												
1	0.52	0.67	0.59												
3	Lasso Regularisation Feature Selection with Balanced Class Weights (9 features)	<table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td></tr><tr><td>0</td><td>0.86</td><td>0.75</td><td>0.80</td></tr><tr><td>1</td><td>0.50</td><td>0.67</td><td>0.57</td></tr></table>		precision	recall	f1-score	0	0.86	0.75	0.80	1	0.50	0.67	0.57	Train AUC: 0.72 Test AUC: 0.78
	precision	recall	f1-score												
0	0.86	0.75	0.80												
1	0.50	0.67	0.57												
4	Ridge Regression	<table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td></tr><tr><td>0</td><td>0.80</td><td>0.86</td><td>0.83</td></tr><tr><td>1</td><td>0.54</td><td>0.42</td><td>0.47</td></tr></table>		precision	recall	f1-score	0	0.80	0.86	0.83	1	0.54	0.42	0.47	Train AUC: 0.73 Test AUC: 0.79 (0.788)
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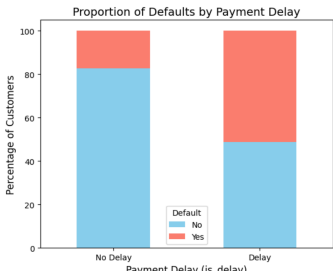
AUC is used as the main evaluation metric because it measures the model's ability to distinguish between defaulting and non-defaulting customers across all classification thresholds, making it robust to class imbalance. This allows for a fair assessment even when defaults are less frequent, as in our dataset (433 non-defaults and 167 defaults). Based on AUC, Models 1, 2, and 3 achieved the highest score of 0.79, with Model 2 slightly outperforming the baseline model (AUC = 0.788). Overall, this indicates that the models can correctly distinguish defaulters from non-defaulters about 79% of the time. To break the tie between Models 2 and 4, we examine precision and recall from the confusion matrices. Since missing defaults (false negatives) is most critical in credit risk, recall for class 1 is the key secondary metric. Model 2 records the highest recall for class 1 (0.67), meaning **67% of borrowers who will default are correctly flagged**. Minimizing false positives is also important to maintain customer relationship. Model 2 achieves the highest precision for class 0 (0.86), **meaning 86% of borrowers predicted as low risk truly do not default**. Therefore, **model 2 with 12 predictors is the recommended model**.

Insights

From the chosen Model 2, the selected predictors provide insights into the most significant drivers of credit default. The coefficients indicate whether each factor increases or reduces the risk of default; their magnitudes reflect the relative strength of that impact. Meanwhile, the odds ratios quantify how strongly each variable influences the likelihood of default. Together, these measures allow us to identify the top predictors that heighten or mitigate default risk.

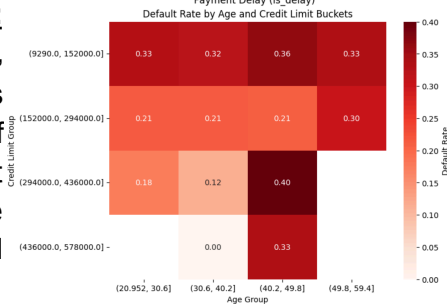


1. Any delays in the past 3 months and the timeliness of repayment in the most recent month increase the risk of default: The higher the value of `is_delay` or `payment_status_1`, that is, the more months a repayment is delayed, the greater the likelihood that a client will default in the following month. This is consistent with the distribution plot, which shows that customers with higher payment status values tend to default more frequently.



2. The combined effect of age and credit limit increases default risk:

The interaction term `age x credit_limit` has a positive coefficient, indicating that being older and having a higher credit limit together increases the likelihood of default, consistent with the darker regions toward the right of the chart. While a higher credit limit alone generally reduces risk, this effect diminishes with age. For older borrowers, larger credit limits may indicate slower repayment behavior or higher exposure, leading to a higher predicted probability of default.



3. Outstanding balances are early warning indicators: The higher `due_amt3`, the higher the risk of default, indicating that unpaid amounts signal deteriorating repayment capacity.

Recommendations

1. Clients with recent payment delays and high outstanding balances should be closely monitored, with **early interventions** such as flexible repayment plans or temporary relief measures to prevent full default.
2. For older clients, credit limit adjustments should be evaluated more cautiously. Rather than applying blanket limits, **risk should be assessed across different credit limit categories**, focusing monitoring efforts on the age-credit limit combinations associated with higher predicted default risk.